THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** lie in the same plane	All three are coplanar.
2	**(b)** rotation of the Earth on its axis	Earth spins from west to east.
3	**(a)** living and non-living components interacting with each other	Biotic and abiotic parts together.
4	**(b)** water, air and a suitable temperature	Conditions needed for life are present.
5	**(d)** diffuse reflection	Irregular.
6	**(c)** 27 days	Orbital period of the Moon.
7	**(b)** food chain	Linear sequence.
8	**(d)** nitrogen	About 78 per cent.
9	**(a)** R = 2f	Focus is midway.
10	**(a)** solar calendar	Gregorian.
11	**(d)** decomposition	Recycling.
12	**(b)** a small part of the total water	About 3 per cent is fresh and little is accessible.
13	**(c)** angle of incidence equals angle of reflection	i = r.
14	**(c)** 24 hours	One day.
15	**(d)** sunlight	Non-living.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Incident ray, reflected ray and normal lie in the same plane (1); Angle of incidence equals angle of reflection (1).	1+1
17	Earth rotates from west to east (1); So the Sun appears to move east to west (1).	1+1
18	Community of living organisms interacting with their physical environment (1); Pond, forest (1).	1+1
19	Crust (1); Mantle and core (1).	1+1
20	Left appears as right in the mirror (1); So drivers can read it correctly in the rear-view mirror (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Incident ray strikes mirror at the point O (1); Normal drawn at O (1); Reflected ray makes angle r = i on the other side (1).	1+1+1
22	Fix a vertical stick on level ground in the Sun (1); Mark the end of its shadow every hour (1); Shadow positions show the time of the day (1).	1+1+1
23	Biotic: algae, fish, frogs, bacteria (1); Abiotic: water, light, temperature, minerals (1); They interact with each other (1).	1+1+1
24	Crust: thin and solid (1); Mantle: thick, hot, semi-molten rock (1); Core: outer liquid, inner solid, iron and nickel (1).	1+1+1
	OR: Reflecting surface curves inwards vs outwards (1); Concave converges rays; convex diverges (1); Concave can form real images; convex only virtual (1).	1+1+1
25	Earth's axis is tilted by 23.5 degrees (1); The tilt keeps the same direction as Earth revolves (1); Sun's rays fall differently, changing day length (1).	1+1+1
	OR: Bacteria convert atmospheric nitrogen to usable form (1); Rhizobium in root nodules (1); Plants use it to make proteins (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Incident and reflected ray on same side of normal (1); i = r (1); Coplanar (1); Activity: pins and protractor on paper (1); Measured angles are equal (1).	1+1+1+1+1

	OR: Rotation: day and night (1); Revolution: year (1); Axis tilt 23.5 degrees (1); Seasons (1); Day length changes (1).	1+1+1+1+1
27	Biotic: producers, consumers, decomposers (1); Abiotic: light, water, air, soil (1); Producers need abiotic factors (1); Consumers depend on producers (1); Decomposers recycle matter (1).	1+1+1+1+1
	OR: Crust: solid, thin (1); Mantle: thick, hot rock (1); Outer core: liquid iron and nickel (1); Inner core: solid (1); Studied through earthquake waves (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** crust	Thin solid layer.
2	**(d)** regular reflection	Parallel beam stays parallel.
3	**(a)** 29.5 days	Synodic period.
4	**(d)** grass, grasshopper, frog, snake	Producer first.
5	**(a)** 21 per cent	One-fifth.
6	**(c)** convex	Wide field of view.
7	**(c)** lunar calendar	Months follow the Moon.
8	**(c)** carbon dioxide	CO2 + water.
9	**(c)** ice caps and glaciers	Frozen.
10	**(a)** 35 degrees	i = r.
11	**(d)** 365 and a quarter days	One year.
12	**(a)** bacteria	Living.
13	**(c)** mantle	Thickest layer.
14	**(b)** reflects light into our eyes	Reflection.
15	**(b)** full moon	Purnima.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Sun is at its highest point in the sky (1); Its rays fall almost vertically (1).	1+1
17	Break down dead matter and recycle nutrients (1); Bacteria and fungi (1).	1+1
18	It supports life (1); It has water, air and suitable temperature (1).	1+1
19	Convex lens is thick in the middle and converges light (1); Concave lens is thin in the middle and diverges light (1).	1+1
20	Solar calendar (1); Lunar calendar (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Not too hot or too cold (1); Water remains liquid (1); Habitable zone (1).	1+1+1
22	Spectacles (1); Cameras (1); Magnifying glass or microscope (1).	1+1+1
23	Day: one rotation of Earth (1); Month: one phase cycle of the Moon (1); Year: one revolution around the Sun (1).	1+1+1
24	Break down dead plants and animals (1); Return nutrients to soil (1); Keep surroundings clean (1).	1+1+1
	OR: Earth has an atmosphere; the Moon has none (1); Earth has liquid water (1); Moon has extreme temperatures (1).	1+1+1
25	Angle with the normal = 90 - 40 (1); i = 50 degrees (1); r = 50 degrees (1).	1+1+1
	OR: Morning: long, towards west (1); Noon: shortest (1); Evening: long, towards east (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Convex converges (1); Concave diverges (1); Focus and focal length (1); Convex: magnifier, camera (1); Concave: spectacles for short sight (1).	1+1+1+1+1
	OR: Sunlit half always faces the Sun (1); Moon orbits the Earth (1); Angle between Sun and Moon changes (1); We see varying fraction of the lit half (1); Full moon when Earth is between (1).	1+1+1+1+1
27	Food web is disturbed (1); Prey populations rise (1); Vultures: carcasses are not cleaned (1); Disease spreads (1); Balance is lost (1).	1+1+1+1+1

	OR: Absorbs UV (1); CFCs destroy ozone (1); Ozone hole over Antarctica (1); Montreal Protocol banned CFCs (1); Recovery is underway (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** producers	Green plants.
2	**(b)** troposphere	Where we live.
3	**(b)** concave mirror	Magnified upright image.
4	**(c)** 365 days (366 in a leap year)	Solar.
5	**(c)** oxygen	Oxygen.
6	**(c)** groundwater	Stored underground.
7	**(d)** normal	Always from the normal.
8	**(b)** 366 days	An extra day is added to February.
9	**(c)** producers	Autotrophs.
10	**(d)** inner core	Solid, very hot.
11	**(b)** 3	n = 360/90 - 1.
12	**(c)** new moon	Amavasya.
13	**(b)** many interlinked food chains	Network.
14	**(d)** stratosphere	Lies above the troposphere.
15	**(b)** concave mirror	Enlarged image.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Variety of living organisms in an area (1); Keeps the ecosystem stable (1).	1+1
17	Provides oxygen for breathing (1); Protects from harmful rays and temperature extremes (1).	1+1
18	Regular: from smooth surface, parallel beam stays parallel (1); Diffuse: from rough surface, scattered (1).	1+1
19	Rotation of the Earth on its axis (1); Half the Earth faces the Sun at a time (1).	1+1
20	Biotic: plants, animals (1); Abiotic: air, water, soil, sunlight (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Producers make food (1); Consumers eat producers or other consumers (1); Decomposers recycle nutrients (1).	1+1+1
22	Nitrogen about 78 per cent (1); Oxygen about 21 per cent (1); Argon, carbon dioxide and other gases about 1 per cent (1).	1+1+1
23	Pole: centre of the mirror surface (1); Principal axis: line through C and P (1); Focus: where parallel rays meet or appear to meet (1).	1+1+1
24	Day and night (1); Apparent motion of the Sun and stars (1); Time zones (1).	1+1+1
	OR: Loss of habitat (1); Soil erosion (1); Reduction in rainfall and rise in CO2 (1).	1+1+1
25	Most water is in oceans and is salty (1); Most fresh water is frozen in ice caps (1); Little is in rivers and lakes (1).	1+1+1
	OR: Parallel rays converge to the focus (1); A bright spot forms on the screen (1); Image is real and inverted (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Virtual, erect, same size (1); Laterally inverted (1); Image distance = object distance (1); n = 360/90 - 1 = 3 (1); Images formed by multiple reflections (1).	1+1+1+1+1
	OR: New moon (1); Waxing crescent, first quarter, gibbous (1); Full moon (1); Waning phases (1); Reflected sunlight and relative positions (1).	1+1+1+1+1

27	Chain: grass, deer, tiger (1); Chain: algae, small fish, big fish (1); Web has many links (1); If one organism is lost, alternatives exist (1); Energy passes at each trophic level (1).	1+1+1+1+1
	OR: Troposphere: weather, life (1); Stratosphere: ozone layer (1); Mesosphere: meteors burn (1); Thermosphere: very hot, auroras (1); Exosphere: merges with space (1).	1+1+1+1+1

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